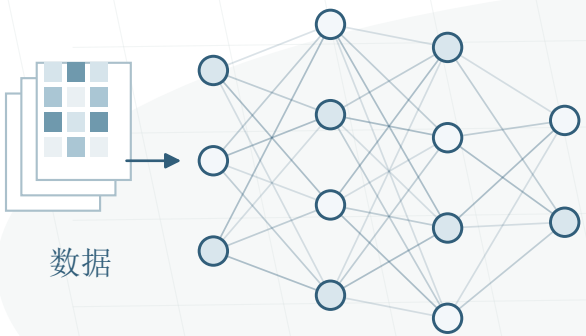
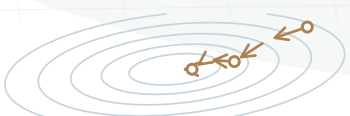
## 学习表示

$$h^{(l+1)} = \sigma(W^{(l)} h^{(l)} + b^{(l)})$$



数据

神经网络



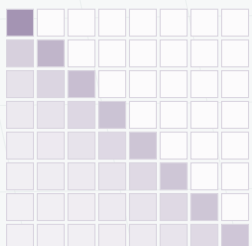
$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

## 建立关联

表示

$Q$

$K$



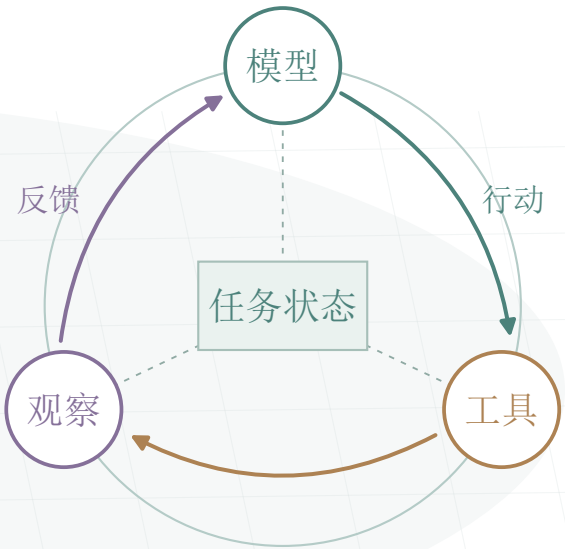
因果注意力

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)$$

上下文 · 关联 · 选择

## 组织行动

上下文




AI Agent

$$s_{t+1} = F(s_t, a_t, o_{t+1})$$